

Email Reader System (Laravel + n8n + MySQL + phpMyAdmin)

Complete Docker setup guide for reusable projects.

Project Goal

Build a Docker-based system using:

- Laravel backend
- n8n automation platform
- Shared MySQL database
- phpMyAdmin
- Email processing workflow

Flow:

Email → n8n → Laravel API → MySQL

Step 1: Create Project Structure

Create project:

```
mkdir "AI Email"
cd "AI Email"
```

Create folders:

```
mkdir laravel-app
mkdir mysql
mkdir mysql\init
mkdir n8n
```

Structure:

```
AI Email/
|
```

```
|─ laravel-app/  
|─ mysql/  
|   └─ init/  
|─ n8n/  
|─ docker-compose.yml  
└─ .env
```

Step 2: Install Laravel

PowerShell:

```
docker run --rm -v "${PWD}/laravel-app:/app" -w /app composer:latest composer  
create-project laravel/laravel .
```

Verify:

```
dir laravel-app
```

Expected:

```
artisan  
app  
bootstrap  
config  
routes  
storage  
vendor
```

Step 3: Create Root .env

Create:

```
New-Item .env
```

Add:

```
MYSQL_ROOT_PASSWORD=root123
MYSQL_DATABASE=email_reader
MYSQL_USER=appuser
MYSQL_PASSWORD=apppass
```

```
N8N_USER=admin
N8N_PASSWORD=admin123
```

```
APP_PORT=8000
N8N_PORT=5678
MYSQL_PORT=3306
PHPMYADMIN_PORT=8080
```

Step 4: Create Laravel Dockerfile

Create:

laravel-app/Dockerfile

Add:

```
FROM php:8.3-cli

RUN apt-get update && apt-get install -y
    git
    unzip
    zip
    curl
    libzip-dev
    default-mysql-client

RUN docker-php-ext-install pdo pdo_mysql zip

COPY --from=composer:latest /usr/bin/composer /usr/bin/composer

WORKDIR /var/www

COPY . .

RUN composer install --ignore-platform-reqs

RUN chmod -R 777 storage bootstrap/cache
```

```
EXPOSE 8000
```

```
CMD sh -c "php artisan key:generate && php artisan serve --host=0.0.0.0 --port=8000"
```

Step 5: Configure Laravel Database

Open:

laravel-app/.env

Replace DB section:

```
DB_CONNECTION=mysql
DB_HOST=mysql
DB_PORT=3306
DB_DATABASE=email_reader
DB_USERNAME=appuser
DB_PASSWORD=apppass
```

Step 6: Create docker-compose.yml

Create root docker-compose.yml

```
services:

  mysql:
    image: mysql:8
    container_name: email_mysql
    restart: unless-stopped

    environment:
      MYSQL_ROOT_PASSWORD: ${MYSQL_ROOT_PASSWORD}
      MYSQL_DATABASE: ${MYSQL_DATABASE}
      MYSQL_USER: ${MYSQL_USER}
      MYSQL_PASSWORD: ${MYSQL_PASSWORD}

    ports:
      - "${MYSQL_PORT}:3306"
```

```
volumes:
  - mysql_data:/var/lib/mysql
  - ./mysql/init:/docker-entrypoint-initdb.d

networks:
  - appnetwork

phpmyadmin:
  image: phpmyadmin/phpmyadmin
  container_name: phpmyadmin

  restart: unless-stopped

  ports:
    - "${PHPMYADMIN_PORT}:80"

  environment:
    PMA_HOST: mysql
    PMA_PORT: 3306
    MYSQL_ROOT_PASSWORD: ${MYSQL_ROOT_PASSWORD}

  depends_on:
    - mysql

  networks:
    - appnetwork

laravel:
  build:
    context: ./laravel-app

  container_name: laravel_app

  ports:
    - "${APP_PORT}:8000"

  volumes:
    - ./laravel-app:/var/www

  depends_on:
    - mysql

  networks:
    - appnetwork
```

```

n8n:
  image: n8nio/n8n:latest

  container_name: n8n_app

  ports:
    - "${N8N_PORT}:5678"

  environment:
    DB_TYPE: mysqlldb
    DB_MYSQLDB_HOST: mysql
    DB_MYSQLDB_DATABASE: ${MYSQL_DATABASE}
    DB_MYSQLDB_USER: ${MYSQL_USER}
    DB_MYSQLDB_PASSWORD: ${MYSQL_PASSWORD}

    N8N_BASIC_AUTH_ACTIVE: true
    N8N_BASIC_AUTH_USER: ${N8N_USER}
    N8N_BASIC_AUTH_PASSWORD: ${N8N_PASSWORD}

  volumes:
    - n8n_data:/home/node/.n8n

  depends_on:
    - mysql

  networks:
    - appnetwork

volumes:
  mysql_data:
  n8n_data:

networks:
  appnetwork:

```

Step 7: Start System

```

docker compose build --no-cache
docker compose up -d

```

Check:

```
docker ps
```

Step 8: Access URLs

Laravel:

<http://localhost:8000>

n8n:

<http://localhost:5678>

phpMyAdmin:

<http://localhost:8080>

n8n Login:

admin admin123

MySQL:

root root123

Step 9: Create Email API

Create model:

```
docker exec -it laravel_app php artisan make:model Email -m
```

Controller:

```
docker exec -it laravel_app php artisan make:controller EmailController
```

API Route:

routes/api.php

```
Route::post('/email/store',[EmailController::class,'store']);
```

Controller:

```
public function store(Request $request)
{
    Email::create([
        'sender'=>$request->sender,
        'subject'=>$request->subject,
        'body'=>$request->body
    ]);

    return response()->json([
        'success'=>true
    ]);
}
```

Step 10: n8n Workflow

Workflow:

IMAP Trigger ↓ Read Email ↓ HTTP Request ↓ Laravel API ↓ MySQL

POST:

<http://laravel:8000/api/email/store>

JSON:

```
{
  "sender": "{{${json.from}}",
  "subject": "{{${json.subject}}",
  "body": "{{${json.text}}",
}
```

Daily Commands

Start:


```
docker compose up -d
```

Stop:

```
docker compose down
```

Rebuild:

```
docker compose build --no-cache
```

Logs:

```
docker logs laravel_app
```